coursera | UNIVERSITY OF MICHIGAN

Applied Information Extraction in Python

Today's goal • 3/3 items

Module 1
Information Extraction

Module 2
Named Entity Recognition (NER)

Named Entities and Named Entity Recognition

- What is Named Entity Recognition (NER)? Video • 5 min
- Knowledge Check: Named Entities and Named Entity Recognition Practice Assignment • Grade: 100%

Setting up NER as a Machine Learning Task

- BIO Encoding for Named Entity Labels Reading • 10 min
- BILOU Encoding for Named Entity Labels Reading • 10 min
- **Machine Learning Fundamentals: How Machines Learn to Label Named Entities** Reading • 10 min
- NER as a Sequence Classification Task Video • 8 min
- Knowledge Check: Setting up NER as a Machine Learning Task Practice Assignment • 10 min

Hidden Markov Models (HMMs)

- Markov Chain and Hidden Markov Models Reading • 10 min
- Fundamentals of Markov Chain Models Video • 7 min
- Hidden Markov Models (HMMs) Video • 11 min
- Training Hidden Markov Models: How HMMs Learn to Assign Labels Reading • 10 min
- The Math Behind HMMs: How Probabilities Power Sequence Labeling Reading • 10 min
- Conditional Random Fields (CRFs) Video • 6 min
- Knowledge Check: Hidden Markov Models (HMMs) Practice Assignment • 15 min

How to Extract Information Using CRFs in Python?

- Demo: CRF Model Training Video • 8 min
- Jupyter Notebook Practice on Training CRFs Lab • 1h

Module Wrap-Up & Assignment

- Assignment 2 Introduction: Implementing a CRF Model Video • 4 min
- Build an Information Extraction Pipeline for CRF Based Extraction Programming Assignment • 3h
- Module 2 Assignment Graded Assignment • 30 min

Module 3
Neural Network Models

Module 4
Transformers Transform Information Extraction

Machine Learning Fundamentals: How Machines Learn to Label Named Entities

0:17 / 1:39 2x

From Labels to Learning

We explored how structured label encodings like BIO and BILOU help us format data for sequence labeling. But knowing where an entity begins or ends is just one part of the task. We also need to teach machines how to assign those labels **accurately** and **at scale**.

That's where **machine learning** comes in.

What Makes NER a Machine Learning Problem?

Named Entity Recognition (NER) is a classic example of a sequence labeling task: each token in a sentence needs a label, and the choice of that label often depends on the surrounding words.

Unlike simple rule-based methods, such as regular expressions, machine learning allows a model to:

- Learn from data, rather than hand-crafted rules
- Capture patterns and dependencies in word sequences
- Adapt to different domains, such as political texts, biomedical records, or clinical notes

Learning from Context

Let's take this sentence: "Dr. Smith visited the University of Michigan on Monday."

A **rule-based system** might flag "Dr." as a title or "Monday" as a day of the week.

But what if the sentence were: "University Drive intersects Smith Street every Monday."

In this case, those same words have **different roles**.

Machine learning models learn from examples rather than fixed rules. They figure out:

- That "Dr." usually precedes a person's name
- That capitalized words at the end of addresses might be streets or cities
- That some words, like "Smith," can be ambiguous

These insights allow the model to generalize, crucial for real-world applications like extracting patient names, medication names, or organizations from messy clinical text.

Mark as completed

Like Dislike Report an issue

Go to next item →